

Requested Procedure: Right pectoral dual-chamber pacemaker implant
Clinical Context: Exertional fatigue and presyncope. Clinic ECG shows bradycardia; Holter confirms high-grade AV block with symptomatic pauses.

1) Vitals and Examination

Measure	Value	Reference Range	Status
Blood Pressure (sitting)	148/92 mmHg	90-129 / 60-84 mmHg	High
Heart Rate	38 bpm	60-100 bpm	Low
Respiratory Rate	18 /min	12-20 /min	Normal
SpO2 (RA)	97%	>= 95%	Normal
BMI	27.8 kg/m^2	18.5-24.9 kg/m^2	Slightly High

Interpretation: Elevated blood pressure and marked bradycardia at rest are consistent with symptomatic conduction disease.

2) Hematology

Test	Value	Reference Range	Status
Hemoglobin	13.6 g/dL	13.5-17.5 g/dL	Normal
Hematocrit	41%	41-53%	Normal
WBC	6.8 x10^9/L	4.0-11.0 x10^9/L	Normal
Platelets	232 x10^9/L	150-400 x10^9/L	Normal

3) Chemistry and Renal

Test	Value	Reference Range	Status
Sodium	139 mmol/L	135-145 mmol/L	Normal
Potassium	4.7 mmol/L	3.5-5.1 mmol/L	Normal
Chloride	103 mmol/L	98-107 mmol/L	Normal
CO2 (Bicarbonate)	24 mmol/L	22-29 mmol/L	Normal
BUN	18 mg/dL	7-20 mg/dL	Normal
Creatinine	1.1 mg/dL	0.74-1.35 mg/dL	Normal
eGFR	>= 60 mL/min/1.73m^2	>= 60 mL/min/1.73m^2	Normal
Glucose (Fasting)	102 mg/dL	70-99 mg/dL	Slightly High

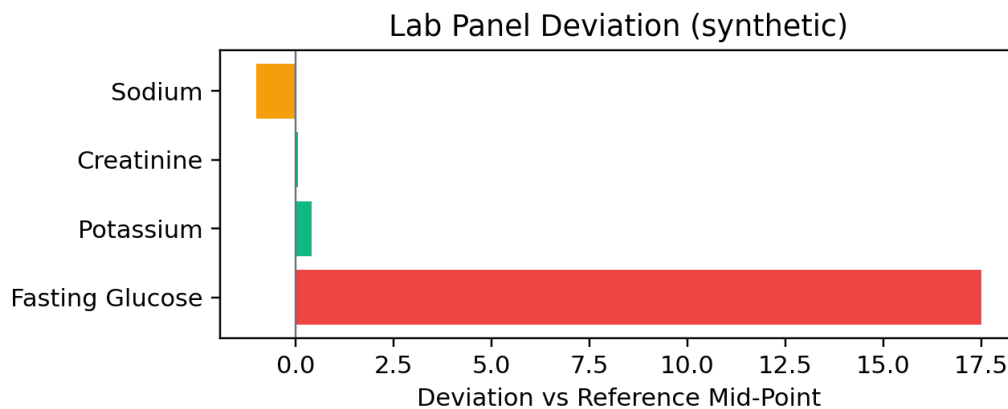


Figure A: Lab panel deviation showing mild elevation of fasting glucose; electrolytes and renal indices within reference limits (synthetic).

4) Coagulation

Test	Value	Reference Range	Status
PT	12.2 sec	11-13.5 sec	Normal
INR	1.0	0.8-1.2	Normal
aPTT	31 sec	25-35 sec	Normal

5) Arrhythmia Findings (Key Reason for Device Therapy)

ECG: Sinus bradycardia with intermittent 2:1 and higher-grade AV block. Holter (24h): mean HR 42 bpm with symptomatic pauses up to 3.1 s; ventricular escape noted during high-grade block. Symptoms (lightheadedness/presyncope) correlate temporally with AV block episodes.

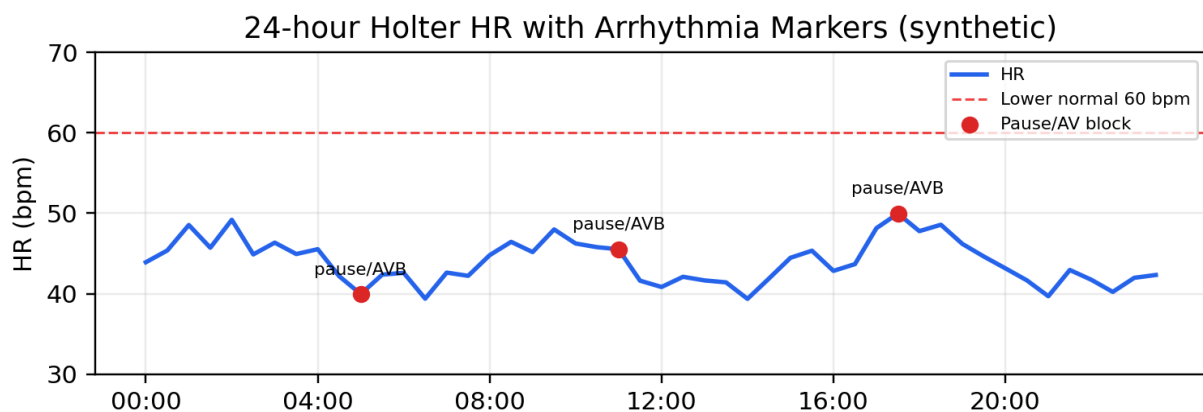


Figure B: 24-hour heart-rate trend with markers at pauses/high-grade AV block; periods below 60 bpm highlight bradycardic burden (synthetic).

6) Why a Pacemaker is Needed (Rhythm-based, not lab-based)

Permanent pacing is indicated due to symptomatic bradyarrhythmia and high-grade atrioventricular (AV) block documented on ECG/Holter, with pauses up to 3.1 seconds and ventricular escape. Normal laboratory values help exclude reversible causes (electrolytes, renal, coagulation) but do not negate conduction disease.

7) Rhythm Abnormalities Summary

7) Rhythm Abnormalities Summary

Finding	Source	Clinical Significance
Resting bradycardia (HR ~38 bpm)	Vitals / ECG	Symptomatic bradycardia; meets pacing criteria when symptomatic.
Intermittent 2:1 AV block	ECG / Holter	Conduction disease with dropped beats; may progress; pacing indicated when symptomatic or high-grade.
High-grade AV block episodes	Holter	Loss of AV conduction leading to instability; pacing is Class I when symptomatic.
Pauses up to 3.1 sec	Holter	Syncope/presyncope risk; pauses > 3 s with symptoms support pacing.
Ventricular escape rhythm	Holter	Backup rhythm indicating advanced conduction system disease.

8) Imaging

Chest X-ray: normal cardiac silhouette; clear lungs. Echocardiogram: LVEF 58%, no significant valvular disease; mild left atrial enlargement.

9) Management Plan and Options

Proceed with right pectoral dual-chamber pacemaker to restore appropriate rate and AV synchrony. Discussed reversible contributors (medications/electrolytes), temporary pacing if needed, and why observation alone is not preferred given symptomatic high-grade block.

10) Clearance

ASA Class II; RCRI low; coagulation within range; no active infection. Cleared for device implantation. Reporting Laboratory: PulseSync Central Diagnostics (demo). Collected 2026-03-17 08:05; received 08:42; reported 12:10.

Disclaimer: Charts and values are synthetic for demonstration and do not constitute medical advice.